



EXAM PAPERS PRACTICE

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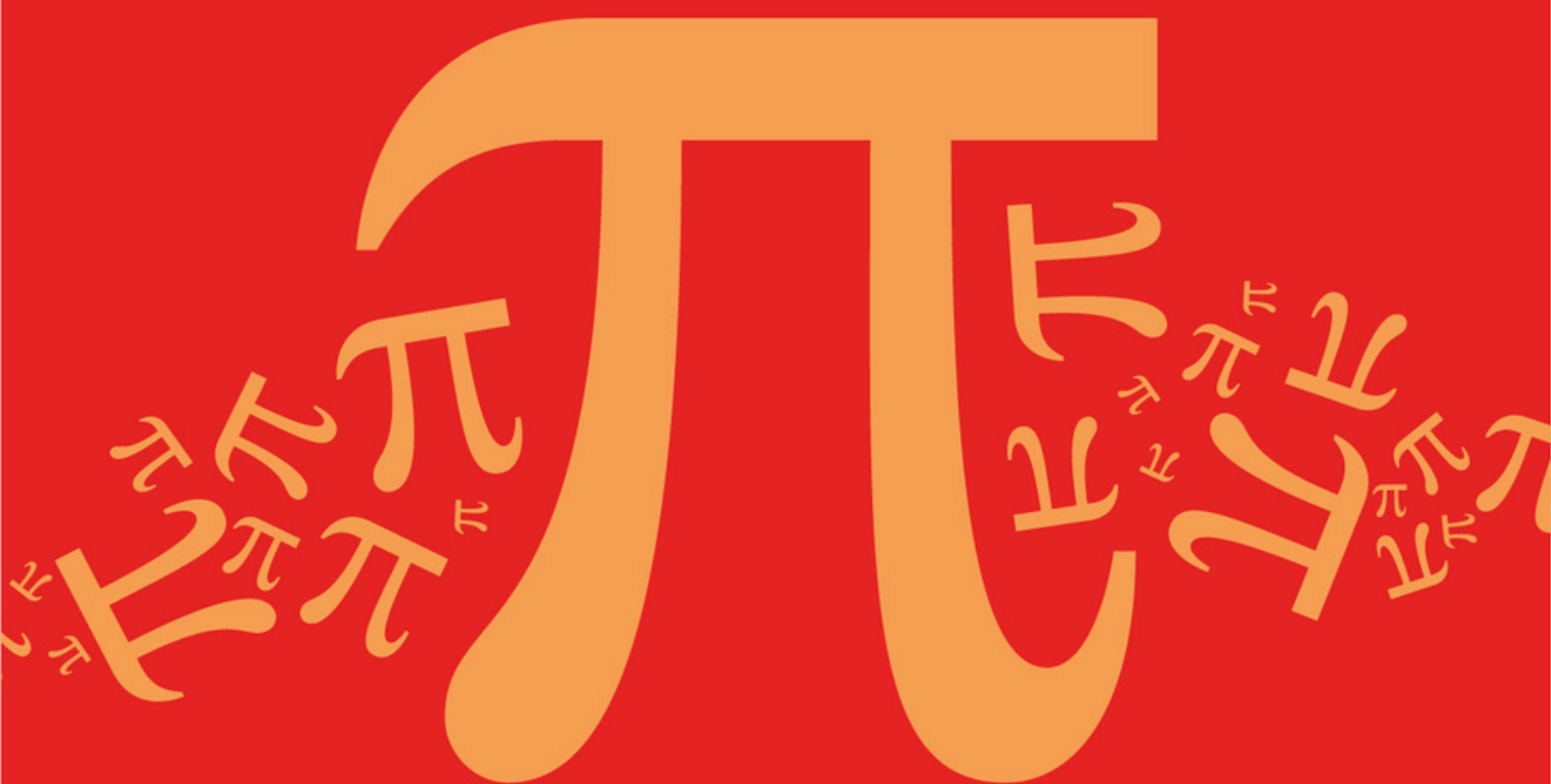
Practice questions created by actual examiners and assessment experts

Detailed mark scheme

Suitable for all boards

Designed to test your ability and thoroughly prepare you

AQA A Level Further Mathematics (7367)



Topic

DA: Graphs

Sub topic

DA4: Kuratowski's Theorem

Topic Questions

AQA A Level Further Mathematics (7367)

Question Paper

Topic

DA: Graphs

Sub topic

DA4: Kuratowski's Theorem

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Name : _____

Date : _____

Time : _____

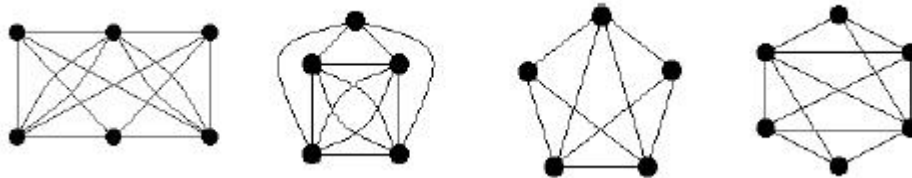
Total Marks Available : _____

Total Marks Received : _____

Q1.

Which of the following graphs is **not** planar?

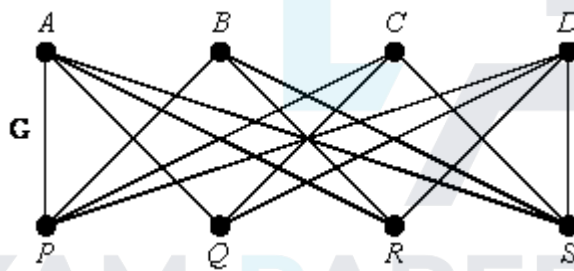
Circle your answer.



(Total 1 mark)

Q2.

The graph **G** shown is the graph $K_{4,4}$.



- (a) Explain what is meant by a Hamiltonian cycle, and give an example from **G**.

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(3)

- (b) How many different Hamiltonian cycles are there in **G**?
(Do not count a cycle taken in reverse order as different from the original.)

(2)

- (c) **G** is non-planar.
Explain what this means.
Use Kuratowski's theorem to prove that **G** is non-planar.

(2)

(Total 7 marks)